

PRESS RECORD

COMPX341 - Advanced Software Engineering Design



THE UNIVERSITY OF
WAIKATO
Te Whare Wānanga o Waikato

Maintenance - Tools and Example

COMPX₃₄₁

Overview

- Maintenance Tools
- Maintenance Example

Maintenance Tools

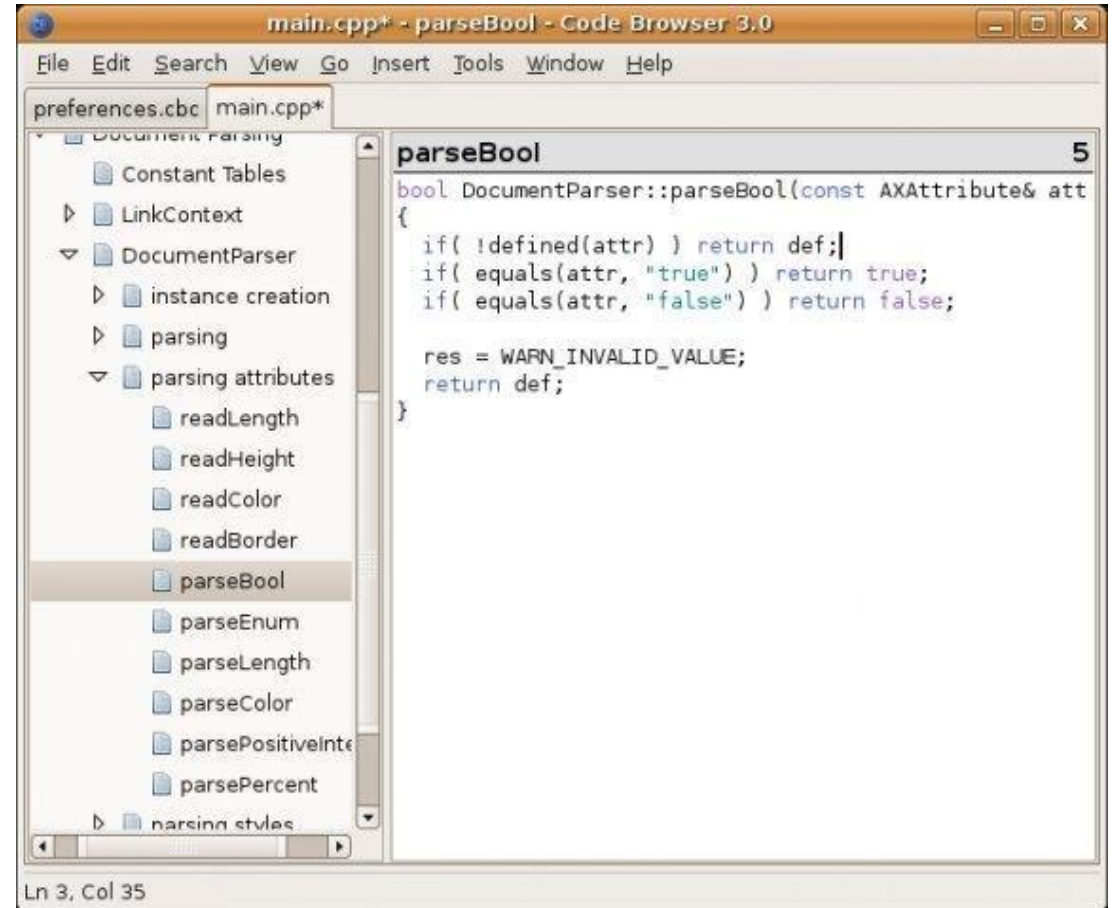
Maintenance Tools

- Code browsers
- Program slicers
- Static analyzers
- Dynamic analyzers
- Data flow analyzers
- ...

Maintenance Tools

Code browser

- A text editor or IDE that includes a folding layout
- Structures source code



Maintenance Tools

Program slicers

- Slicing is the computation of a set of statements
- By creating a slice, you eliminate all other parts of the program that are not currently of interest
- You can then focus on the code that is of interest to you (usually because there is an error occurring)
- Programmer clicks on a statement, then the program slicer tool highlights all statements that might affect the statement of interest (i.e. creates a slice)

Maintenance Tools

Static analyzers

- Static Code Analysis is the analysis of a system without running it.
 - Data flow analysis
 - Control flow graph
 - Taint analysis
 - Lexical analysis
 - ...
- https://owasp.org/www-community/controls/Static_Code_Analysis

Maintenance Tools

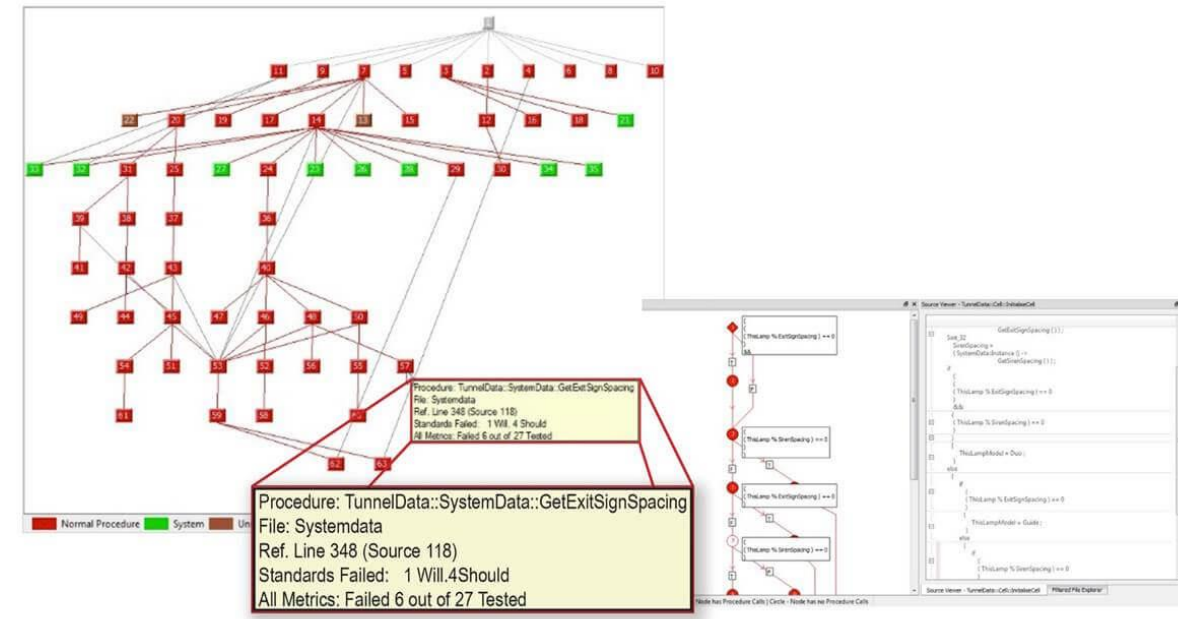
Dynamic analyzers

- Dynamic Code Analysis is the analysis of a system while it is running.
 - Compile time and run time vulnerabilities
 - Configuration errors
- Dynamic Analysis Tools can test
 - SQL queries
 - Long input strings
 - Negative and large positive numbers
 - Unexpected input data
- <https://www.checkpoint.com/cyber-hub/cloud-security/what-is-dynamic-code-analysis/>

Maintenance Tools

Data flow analyzers

- Used to gather information about values that are calculated at different points in a system
 - Control flow graph
 - Setup dataflow equations for each node in the graph
 - Solve them by repeatedly calculating output from given input



- <https://clang.llvm.org/docs/DataFlowAnalysisIntro.html>
- <https://lra.com/capabilities/data-flowcontrol-flow-analysis/>

Maintenance Example

Maintenance Example

"We would like to create a spell checker application. The application will take a word and either confirm that it is spelt correctly, or confirm that it is not spelt correctly. Input words may include punctuation, but cannot include numerical characters"

Maintenance Example

"A user has identified a bug in the code. The input "time()" is expected to result in the output true. Instead, it results in the output false."

"Your company has decided that the system code needs to be re-engineered for greater maintainability"

"The client would like you to add the ability to handle strings with numeric values"

Maintenance Example

- For each:
 1. Problem/modification request
 2. Problem/modification analysis
 - Determine the maintenance category (Preventative, perfective, corrective, adaptive)
 - Conduct an *informal* Impact Analysis evaluation
 3. Modification request acceptance/rejection
 - Determine whether to accept the request
 4. Modification implementation
 - Implement the modification
 5. Maintenance review / acceptance
 - Testing and/or code review

Maintenance Example

"A user has identified a bug in the code. The input "time()" is expected to result in the output true. Instead, it results in the output false."

Maintenance Example

1. Problem/modification request

A user has identified a bug in the code. The input "time()" is expected to result in the output true. Instead, it results in the output false.

Problem request: investigate and fix the bug that results in the input "time()" returning false.

2. Problem/modification analysis

Maintenance category: ?

... remember

Software Maintenance Categories

	Correction Same requirements	Enhancement Changes requirements and/or changed environment
Proactive To prevent failure	Preventive Bugs are there but not surfacing ... yet!	Perfective New features or documentation; better performance; etc.
Reactive Because of one or more reported failures	Corrective a.k.a. Repairs Special case: emergency	Adaptive Changed/changing environment (e.g., updated OS, middleware, etc.)

Maintenance Example

1. Problem/modification request

A user has identified a bug in the code. The input "time()" is expected to result in the output true. Instead, it results in the output false.

Problem request: investigate and fix the bug that results in the input "time()" returning false.

2. Problem/modification analysis

Maintenance category: Corrective

Maintenance Example

1. Problem/modification request

A user has identified a bug in the code. The input "time()" is expected to result in the output true. Instead, it results in the output false.

Problem request: investigate and fix the bug that results in the input "time()" returning false.

2. Problem/modification analysis

Maintenance category: Corrective

Impact Analysis:

- Verify the problem:
- Implementation option:
- Effort:
- Resources:

Maintenance Example

1. Problem/modification request

A user has identified a bug in the code. The input "time()" is expected to result in the output true. Instead, it results in the output false.

Problem request: investigate and fix the bug that results in the input "time()" returning false.

2. Problem/modification analysis

Maintenance category: Corrective

Impact Analysis:

- Verify the problem:
- Implementation option:
- Effort:
- Resources:

Let's take a look ...

Maintenance Example

1. Problem/modification request

A user has identified a bug in the code. The input "time()" is expected to result in the output true. Instead, it results in the output false.

Problem request: investigate and fix the bug that results in the input "time()" returning false.

2. Problem/modification analysis

Maintenance category: Corrective

Impact Analysis:

- Verify the problem: only a subset of punctuation is currently covered
- Implementation option: use *punct* regex to cover a larger group of punctuation characters
- Effort: alter one line of code, no new knowledge required
- Resources: less than 15 minutes of a maintainer's time (what about re-running tests? etc.)

Maintenance Example

3. Modification request acceptance/rejection

Decision: Accepted

4. Modification implementation

- Implement the modification

5. Maintenance review / acceptance

- Re-run regression tests
- Re-run acceptance tests
- Submit to code review
- On approval, submit to Git

Maintenance Example

"Your company has decided that the system code needs to be re-engineered for greater maintainability"

Maintenance Example

1. Problem/modification request

Your company has decided that the system code needs to be re-engineered for greater maintainability

Modification request: Use code and solution quality techniques to modularize the code for better maintainability.

2. Problem/modification analysis

Maintenance category: ?

... remember

Software Maintenance Categories

	Correction Same requirements	Enhancement Changes requirements and/or changed environment
Proactive To prevent failure	Preventive Bugs are there but not surfacing ... yet!	Perfective New features or documentation; better performance; etc.
Reactive Because of one or more reported failures	Corrective a.k.a. Repairs Special case: emergency	Adaptive Changed/changing environment (e.g., updated OS, middleware, etc.)

Maintenance Example

1. Problem/modification request

Your company has decided that the system code needs to be re-engineered for greater maintainability

Modification request: Use code and solution quality techniques to modularize the code for better maintainability.

2. Problem/modification analysis

Maintenance category: Preventative

Maintenance Example

1. Problem/modification request

Your company has decided that the system code needs to be re-engineered for greater maintainability

Modification request: Use code and solution quality techniques to modularize the code for better maintainability.

2. Problem/modification analysis

Maintenance category: Preventative

Impact Analysis:

- Verify the problem:
- Implementation option:
- Effort:
- Resources:

Maintenance Example

1. Problem/modification request

Your company has decided that the system code needs to be re-engineered for greater maintainability

Modification request: Use code and solution quality techniques to modularize the code for better maintainability.

2. Problem/modification analysis

Maintenance category: Preventative

Impact Analysis:

- Verify the problem:
- Implementation option:
- Effort:
- Resources:

Let's take a look ...

Maintenance Example

1. Problem/modification request

Your company has decided that the system code needs to be re-engineered for greater maintainability

Modification request: Use code and solution quality techniques to modularize the code for better maintainability.

2. Problem/modification analysis

Maintenance category: Preventative

Impact Analysis:

- Verify the problem: Code is contained in one class. Should design a new modularized solution.
- Implementation option: re-engineer code to adhere to newly created component and class diagram
- Effort: alter all lines of code, new knowledge?
- Resources: significant amount of the maintainer's time, a full re-engineer of the entire system, plus re-running of all tests. May need a staged approach so as to leave the system live during update.

Maintenance Example

3. Modification request acceptance/rejection

Decision: ???

4. Modification implementation

- Implement the modification

5. Maintenance review / acceptance

- Re-run regression tests
- Re-run acceptance tests
- Submit to code review
- On approval, submit to Git

Maintenance Example

"The client would like you to add the ability to handle strings with numeric values"

Maintenance Example

1. Problem/modification request

The client would like you to add the ability to handle strings with numeric values

Modification request: Added feature. The system must have the ability to strip numeric values from strings.

2. Problem/modification analysis

Maintenance category: ?

... remember

Software Maintenance Categories

	Correction Same requirements	Enhancement Changes requirements and/or changed environment
Proactive To prevent failure	Preventive Bugs are there but not surfacing ... yet!	Perfective New features or documentation; better performance; etc.
Reactive Because of one or more reported failures	Corrective a.k.a. Repairs Special case: emergency	Adaptive Changed/changing environment (e.g., updated OS, middleware, etc.)

Maintenance Example

1. Problem/modification request

The client would like you to add the ability to handle strings with numeric values

Modification request: Added feature. The system must have the ability to strip numeric values from strings.

2. Problem/modification analysis

Maintenance category: Perfective

Maintenance Example

1. Problem/modification request

The client would like you to add the ability to handle strings with numeric values

Modification request: Added feature. The system must have the ability to strip numeric values from strings.

2. Problem/modification analysis

Maintenance category: Perfective

Impact Analysis:

- Verify the problem:
- Implementation option:
- Effort:
- Resources:

Maintenance Example

1. Problem/modification request

The client would like you to add the ability to handle strings with numeric values

Modification request: Added feature. The system must have the ability to strip numeric values from strings.

2. Problem/modification analysis

Maintenance category: Perfective

Impact Analysis:

- Verify the problem:
- Implementation option:
- Effort:
- Resources:

Let's take a look ...

Maintenance Example

1. Problem/modification request

The client would like you to add the ability to handle strings with numeric values

Modification request: Added feature. The system must have the ability to strip numeric values from strings.

2. Problem/modification analysis

Maintenance category: Perfective

Impact Analysis:

- Verify the problem: the system currently disregards strings with numeric values. Instead, the new feature will allow the system to strip numeric values from strings.
- Implementation option: create a new method called stripNumeric that will strip numeric values from a string. Update the source code to use this method rather than disregarding strings with numeric values
- Effort: alter 5-10 lines of code, any new knowledge?
- Resources: less than an hour of a maintainer's time (what about re-running tests? etc.)

Maintenance Example

3. Modification request acceptance/rejection

Decision: Accepted

4. Modification implementation

- Implement the modification

5. Maintenance review / acceptance

- Re-run regression tests
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